Four alternatives are given for each of the following questions / incomplete statements. Only one of them is correct or most appropriate. Choose the correct alternative and write the complete answer along with its letter of alphabet.

$10 \times 1 = 10$

1. The energy conversion based on the principle of photovoltaic effect is

   (A) solar energy into heat energy

   (B) solar energy into mechanical energy

   (C) solar energy into electrical energy

   (D) wind energy into electrical energy.

2. The atomic number of an element '*X*' is 16. In the modern periodic table the element '*X*' belongs to this block and period.

   (A) *P*-block, 3rd period

   (B) *P*-block, 2nd period

   (C) *S*-block, 3rd period

   (D) *S*-block, 2nd period.

3. Use of detergents is hazardous to the aquatic life because, they

   (A) reduce the amount of dissolved oxygen content in water.

   (B) increase the temperature of water body.

   (C) are made from vegetable oils.

   (D) contain heavy metals.

Use E-Papers, Save Trees
above line hide when print out-www.inyatrust.com

4. The technique used to track aircraft is

(A) Sonar

(B) Scanner

(C) Laser

(D) Radar.

5. An example for polyfunctional compound is

(A) Glycine

(B) Ethanol

(C) Ethanamine

(D) Methanal.

6. A student finds a flower in his school campus and decides that it is the flower of a monocot plant. The reason for his conclusion is

(A) petals are colourful.

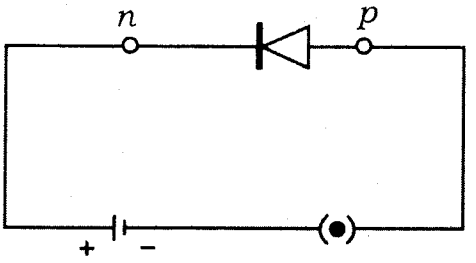
(B) petals are in multiples of four or five.

(C) petals are in multiples of three.

(D) generally flowers are unisexual.

Use E-Papers, Save Trees
above line hide when print out-www.inyatrust.com

7. A diode is connected in a circuit as shown in the figure. The correct statement related to this figure is



(A) diode offers high resistance

(B) diode offers low resistance

(C) maximum charges flow in the circuit

(D) diode offers zero resistance.

8. These diseases can be controlled by eradicating mosquitoes.

(A) Dengue fever, Gonorrhea

(B) Dengue fever, Hepatitis-B

(C) Chikungunya, Syphilis

(D) Chikungunya, Dengue fever.

9. The constituents of alloy which is used in the manufacture of permanent magnets are

(A) Aluminium + Copper + Magnesium + Manganese

(B) Nickel + Cobalt + Iron + Aluminium

(C) Iron + Carbon + Chromium + Nickel

(D) Iron + Carbon + Nickel.

Use E-Papers, Save Trees
above line hide when print out-www.inyatrust.com

10. A special property of DNA that ensures the equal distribution of similar genetic material to the offsprings is

(A) mutation (B) recombination

(C) replication (D) variation.

11. The devices are given in **Column-A** and their uses are given in **Column-B**. Match them and write the answer along with its letters : $4 \times 1 = 4$

Column - A		***Column - B***
(A)	A.C. Dynamo	(i) Stepping up the A.C. voltage to transport electricity to distant places
(B)	Step-up transformer	(ii) In equipment like washing machine work on A.C. source
(C)	D.C. motor	(iii) In discharge tube experiments to obtain very high D.C. voltage from a low D.C. voltage
(D)	Induction coil	(iv) In thermal power station to produce alternate current
		(v) At the distribution point for domestic supply and at the receiving centre to decrease the voltage
		(vi) In vehicles to obtain low voltage D.C. from high voltage A.C.
		(vii) In devices like toys, tape-recorders etc.

Use E-Papers, Save Trees
above line hide when print out-www.inyatrust.com

Answer the following questions : $7 \times 1 = 7$

12. What is geothermal energy ?

13. Write one use of Zeolite.

14. A palaeontologist observes the jaw of an animal and concludes that it belongs to mammals. Give reason for his conclusion.

15. Write one difference between saturated hydrocarbons and unsaturated hydrocarbons.

16. When the mixture of silica and coke is heated in an electrical furnace, silicon carbide is formed instead of silicon. What is the reason ?

17. Mention the function of sympathetic and parasympathetic nervous system with respect to the pupil of eye.

18. State Faraday's first law of electrolysis.

Answer the following questions : $16 \times 2 = 32$

19. Boy *A* argues that light wave is a transverse wave. Boy *B* argues that it is an electromagnetic wave. Whose argument is correct ? Justify your answer scientifically.

20. In the modern periodic table, how does the atomic size of the elements vary along the period and down the group ? Explain.

Use E-Papers, Save Trees
above line hide when print out-www.inyatrust.com

21. The percentage of air pollution from different sources are given in the table. With the help of this, answer the following questions :

Sources of air pollution	*Industry*	*Transportation*	*Consumer & Commercial products*	*Commercial & Residential heating*	*Agriculture*	*Other*
Percentage	52%	27%	8%	1%	10%	2%

(a) Which are the two sources to be monitored more to control air pollution ?

(b) Mention the preventive measure for each one of these two sources to control air pollution.

22. Write any two differences between diesel engine and petrol engine.

OR

Give any four reasons to decide that internal combustion engines are advantageous over steam engines.

23. How can coloured glass be obtained from molten glass ? Name the chemical compounds to be added to molten glass to obtain the following coloured glass.

(a) Yellow glass

(b) Blue glass.

OR

Use E-Papers, Save Trees
above line hide when print out-www.inyatrust.com

Give scientific reason :

(a) Ceramics are used in electrical gadgetry.

(b) Wax paper is used in food preservation.

24. Write the differences between tendons and ligaments.

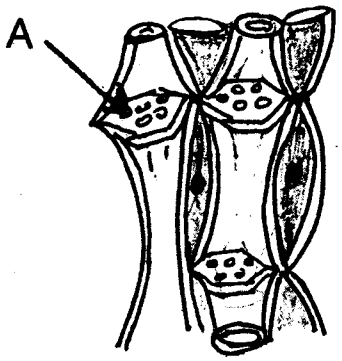
25. A man is standing between hill *A* and hill *B*, claps louder. He hears an echo after 4 seconds from hill *A* and after 6 seconds from hill *B*. The speed of sound in air is 340 ms^{-1}. Calculate the distance between the two hills.

26. The given equation represents the reaction of copper sulphate with an element *X*.

$$CuSO_4 + X \rightarrow Cu + Y$$

Which element is represented by *X*, among Fe and Ag ? Justify your answer. Write the molecular formula of the compound represented by *Y*.

27. The figure of longitudinal section of a plant tissue is given below. Name the part labelled as '*A*' and mention its function. Name the other components of this tissue.



Use E-Papers, Save Trees
above line hide when print out-www.inyatrust.com

28. Draw the diagram showing the expansion stroke of steam engine.

29. Air filled balloon is kept inside the glass jar fitted with a vacuum pump. What will be the change in the size of the balloon when air is taken out with the help of vacuum pump from the jar ? State the law which supports your answer.

30. Draw the diagram showing the structure of HIV.

31. What is superconductivity ? Mention any two uses of superconductors.

OR

What is a transistor ? Mention any two uses of transistor.

32. What are polyploids ? Write their limitations.

OR

Mention the advantages of using antioxidants and colourants in food processing.

33. Draw the diagram of the apparatus used in electroplating.

34. What are the physical changes that resulted in the upright posture in the course of evolution of human ?

OR

Write any four physical features of Australopithecus man.

Use E-Papers, Save Trees
above line hide when print out-www.inyatrust.com

Answer the following questions : $5 \times 3 = 15$

35. Draw the diagram of a nuclear power reactor and label the following parts.

(a) Reflector

(b) Heat exchanger.

36. (a) Write the steps of manufacturing sugar from sugarcane.

(b) In the manufacture of sugar mention the importance of the following :

(i) Norit

(ii) Calcium hydroxide.

OR

Briefly explain the manufacture of ethyl alcohol from molasses.

37. (a) Mention any four flight adaptations in birds.

(b) Name the two major groups of fishes based on the composition of endoskeleton.

38. Draw the diagram of the blast furnace used in the extraction of iron and label the following parts :

(a) Molten iron

(b) Slag.

Use E-Papers, Save Trees
above line hide when print out-www.inyatrust.com

39. Explain the double helix structure of DNA molecule.

OR

Explain the Carl Correns's monohybrid cross in Four O'clock plant with the help of schematic representation. Mention the phenotype ratio and genotype ratio of plants occurred in F_2 generation.

Answer the following questions : $3 \times 4 = 12$

40. (a) Write the evolutionary stages of the sun like star. Explain the last stage.

(b) Why rockets have to carry oxidizer along with the fuel ?

OR

(a) Explain Big Bang theory.

(b) What are Geostationary satellites ? Why are they called communication satellites ?

41. (a) What is isomerism ? Name the isomers of butane.

(b) Name the air pollutant liberated in the incomplete combustion of methane.

(c) Write the importance of hydrogenation of oils.

42. Draw the vertical section of the human eye and label the following parts :

(a) Lens

(b) Fovea.

Use E-Papers, Save Trees
above line hide when print out-www.inyatrust.com